

GITHUB-

1. Delhi Pollution

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new
        Scanner(System.in);

        int N =
        sc.nextInt(); int[]
        a = new int[N];

        for (int i = 0; i < N;
            i++) { a[i] =
                sc.nextInt();
            }

        int D =
        sc.nextInt(); int X
        = sc.nextInt();

        int fine = 0;

        for (int i = 0; i < N;
            i++) { if (D % 2
                == 0) {
                    if (a[i] % 2 !=
                        0) fine +=
                        X;
                } else {
                    if (a[i] % 2 ==
                        0) fine += X;
                }
            }

        System.out.println(fine);
    }
}
```

}

Output

600

2. Ayush String Transformation

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new
        Scanner(System.in);

        int n =
        sc.nextInt();
        String A =
        sc.next(); String
        B = sc.next();

        int operations = 0;

        for (int i = 0; i < n; i++) {
            if (A.charAt(i) <
                B.charAt(i)) {
                System.out.println(-1);
                return;
            }
            if (A.charAt(i) != B.charAt(i)) {
                operations++;
            }
        }

        System.out.println(operations > 0 ? 1 : 0);
    }
}
```

Output

1

3. Jack's Drum Game

```
import

java.util.*;
```

```
public class Main
```

```
{
```

```
    static long gcd(long a, long  
        b) { while (b != 0) {  
        long temp = b;
```

```

        b = a %
        b; a =
        temp;
    }
    return a;
}

```

```

static long lcm(long a, long
    b) { return (a / gcd(a, b)) *
    b;
}

```

```

public static void main(String[] args) {
    Scanner sc = new
    Scanner(System.in);

```

```

    int N = sc.nextInt();
    int[] board = new int[N];

```

```

    for (int i = 0; i < N; i++) {
        board[i] = sc.nextInt() -
        1;
    }

```

```

    boolean[] visited = new
    boolean[N]; long answer = 1;

```

```

    for (int i = 0; i < N;
        i++) { if
        (!visited[i]) {
            int len =
            0; int
            curr = i;

```

```

            while
            (!visited[curr])
            { visited[curr] =
            true; curr =
            board[curr];
            len++;
            }

```

```
        answer = lcm(answer, len);
    }
}

System.out.println(answer);
}
```

Output

6

4.Verbal Kho-Kho

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new
        Scanner(System.in);

        int N =
        sc.nextInt(); int[]
        D = new int[N];

        for (int i = 0; i < N;
            i++) { D[i] =
            sc.nextInt();
        }

        int count = 0;

        for (int i = 1; i < N;
            i++) { if (D[i] !=
            D[i - 1]) {
                count++;
            }
        }

        System.out.println(count);
    }
}
```

Output

4

5. Golden House

```
import java.util.*;

public class Main {
```

```
public static void main(String[] args) {  
    Scanner sc = new  
    Scanner(System.in);  
  
    int N =  
    sc.nextInt(); int K  
    = sc.nextInt();
```



```

int[] arr = new int[N];

for (int i = 0; i < N;
    i++) { arr[i] =
    sc.nextInt();
}

int start = 0, sum = 0;

for (int end = 0; end < N;
    end++) { sum += arr[end];

    while (sum > K) {
        sum -=
        arr[start];
        start++;
    }

    if (sum == K) {
        System.out.println((start + 1) + " " + (end +
        1)); return;
    }
}
}
}

```

Output

1 3

6. Horse Riding

```

import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new
        Scanner(System.in);

        int N =

```

```
sc.nextInt(); int K  
= sc.nextInt();
```

```
int[] arr = new int[N];
```

```
for (int i = 0; i < N; i++) {
```

```
        arr[i] = sc.nextInt();
    }

    int start = 0;
    long sum =
    0; int
    maxLen = 0;

    for (int end = 0; end < N;
        end++) { sum += arr[end];

        while (sum >= K)
            { sum -=
              arr[start];
              start++;
            }

        maxLen = Math.max(maxLen, end - start + 1);
    }

    System.out.println(maxLen);
}
}
```

Output

3